B.PASINDU MANODARA BOLONGHE

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Profile Summary

I am a forward-thinking person driven by a passion for innovation and a deep curiosity about where technology meets creativity. I focus on virtual reality (VR) application development, using it to create immersive experiences that solve real-world problems, while also exploring machine learning and cybersecurity. I welcome challenges as chances to learn and grow.

VR Application Development | Machine Learning | Cybersecurity

Projects & Researches

Motor-Mediated Creativity: Bridging Embodied Skill Training and Digital Expression *2025*

This research introducing motor-mediated creativity, a paradigm that cultivates motor fluency to enhance expressive digital drawing by addressing sensorimotor bottlenecks in novices. Through Motus, an interactive system, we integrated explicit motor training (e.g., velocity- and pressure-modulated strokes) with ideation prompts to reduce cognitive load in translating intent to output, grounded in embodied cognition and motor learning. Formative and evaluation studies demonstrated improved expressive agency and artwork quality, submitted to **CHI2026**.

HCI | Motor Learning | Expressive Motor Control

Competitive virtual reality cycling over low latency networks *2025*

As team leader for our final-year project, I directed the creation of an innovative VR cycling simulator emphasizing seamless navigation and realistic motion dynamics. The system delivers competitive multiplayer racing with low-latency networking for synchronized rider positioning, immersive 360-degree environmental traversal, and photorealistic real-world terrains. Key to our motion fidelity, we integrated adaptive resistance feedback in the initial prototype, simulating dynamic uphill pedaling and downhill momentum to enhance proprioceptive realism. Hosted at the Dialog Innovation Center in the Lotus Tower for imminent public demonstration, my contributions centered on leading the Oculus VR application in Unity 3D, optimizing gesture-based steering for intuitive path navigation and fluid body-motion tracking. Our associated research paper on Virtual Reality Cycling: a Competitive and Low-Latency Exer-Gaming Environment was presented at IEEE Gaming , media & Entertainment (**IEEE GEM2025**) conference 2025.

Unity 3D | c# | Motor Navigation | IoT

Machine learning project *2024*

As a part of my personal interests, I engage in machine learning projects focusing on image and text classifications and model tuning . For a detailed showcase of my work, you can explore my GitHub account.

tensorflow | numpy | pytorch

Working Experience

Singapore Management University *Jan 2026 - Now*
Research Engineer
Modeling the effect of exploration-inducing phenotypes on age-related deficits in urban spatial cognition

Thakshana Technologies (Pvt) Ltd *Jun 2024 - Dec 2025*
Embedded software Engineer
Design & develop APIs, communication protocols, and cryptographic applications for embedded systems

C++ | c | Certificates | Signatures

KATIM Abu Dhabi,UAE *oct 2024, Nov 2025*
Embedded software Engineer
Got a training in KATIM in embedded software engineering

4Axis Solutions (Pvt) Ltd *Jan 2023 - June 2023*
Software Engineering Intern
I was a member of the VR team where I actively contributed to the development of a drawing application named Drawing Desk specifically designed for Oculus VR devices. During my internship, I acquired proficiency in Unity 3D and C#, skills that I continue to apply and enhance through my ongoing commitment to learning, particularly in the context of my final year project. The application which we developed can be found here.

Unity 3D | c# | Unity shaders | Database - firebase

Education

University of Moratuwa Moratuwa, Sri Lanka
B.Sc. Engineering (Hons.) in Electronics and Telecommunications *Dec 2019 - Dec 2024*
CGPA: 3.70 (First class)

D.S Senanayake College Colombo 07, Sri Lanka
GCE Advanced Level - Physical Science Stream 2016-2018
Z-Score 2.1817

Prince Of Wales college Moratuwa, Sri Lanka

Skills Summary

Languages: Python, C++, C#, C, matlab, Arduino
Tools: Unity 3D, Solidworks, Altium
Soft Skills: Leadership, Organizing, Event Management
Technical Skills: OpenCV, tensorflow, numpy, VS code, Git

Achievements and Awards

SPARK Challenge 21/22: Top 10 finalists
IEEE Signal Processing Cup 2022: Worldwide 21st Place
Future Innovators Challenge'22 : Semi finalist

Courses and certificates

DeepLearning.AI TensorFlow Developer Professional Certificate *2022*
Introduction to tensorFlow for artificial intelligence, machine Learning, and deep Learning

DeepLearning.AI Deep Learning Specialization *2023*
Improving deep neural networks: hyperparameter tuning, regularization and optimization

Data science foundations: Data structures and algorithms specialization *2023*
University of Colorado Boulder

HCIA-AI course *2021*
Huawei AI course(HCIA-AI) by Huawei talent Online in University of Moratuwa ICT Academy

Leadership and Volunteering

Rotaract Club - University of Moratuwa *2020-2021*
Co-chairperson of "Andurata Athwelak" computer course project launched by Rotaract Mora.
That project was awarded the most outstanding Rotaract project of the year 2020/2021

Electronic Club - University of Moratuwa *2022-2023*
Social media coordinator

References

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